Name____________________

Dividing Polynomials

Date________________ Period____

Determine if $d(x)$ is a factor of $f(x)$.

1) $f(x) = 3x^4 + 7x^3 + 3x^2 - x - 4$
$d(x) = x + 2$

2) $f(x) = 2x^3 - x^2 - 6x - 1$
$d(x) = x + 1$

3) $f(x) = 4x^3 - 2x^2 + x - 3$
$d(x) = x - 1$

4) $f(x) = 3x^4 - 3x^3 - 9x^2 + 5x - 2$
$d(x) = x - 2$

Divide. Write your answer in fraction form.

5) $(2x^5 - 15x^3 - 9x^2 + 11x + 12) \div (x + 2)$

6) $(x^4 - x^3 - 19x^2 - 3x - 19) \div (x - 5)$

7) $(10x^4 - 4x^3 + 14x^2 - 14x - 16) \div (2x - 2)$

8) $(9x^5 - 9x^4 - x^3 - 12x^2 + x - 11) \div (3x - 5)$